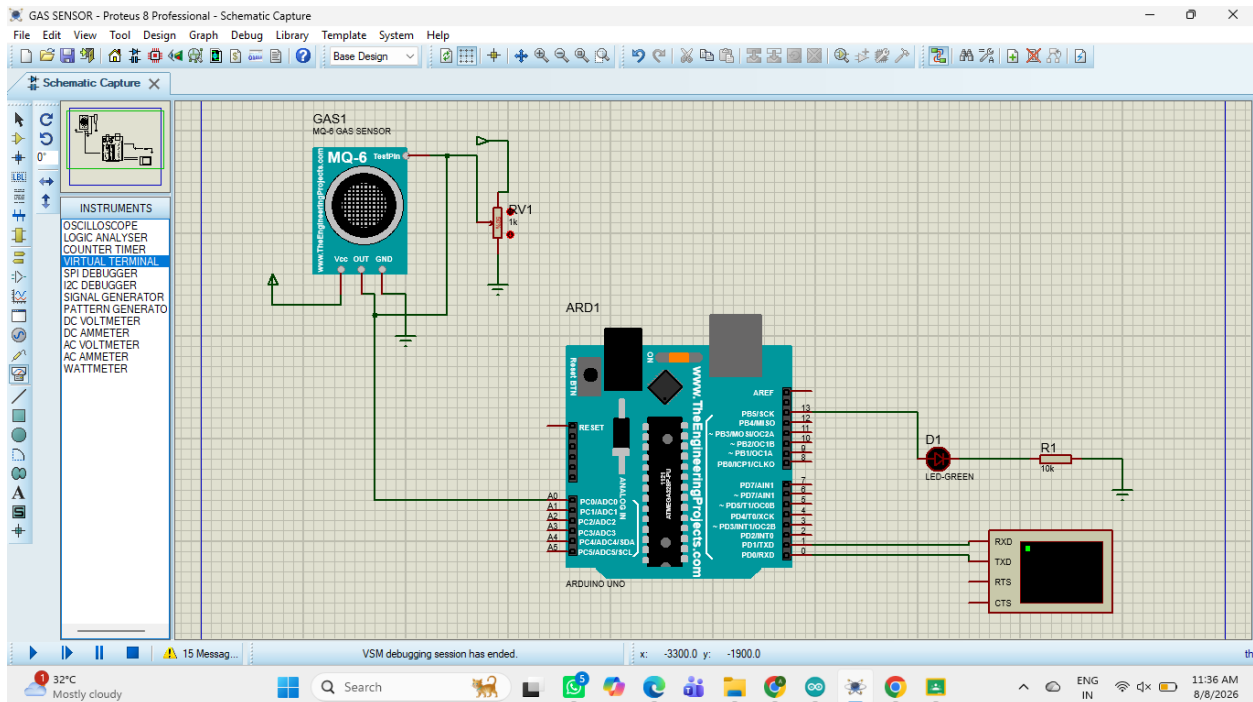


8. GAS SENSOR MQ-2 IN PROTEUS USING ARDUINO



```
exp08 | Arduino 1.8.19
File Edit Sketch Tools Help

exp08

void setup() {
  // Initialize serial communication at 9600 bits per second
  Serial.begin(9600);

  // Set pin modes
  pinMode(mq6Pin, INPUT);
  pinMode(alertPin, OUTPUT);
}

void loop() {
  // Read the analog value from the MQ-6 sensor (returns 0 to 1023)
  int sensorValue = analogRead(mq6Pin);

  // Print the data to the Serial Monitor / Virtual Terminal
  Serial.print("MQ6 Sensor Value: ");
  Serial.println(sensorValue);

  // Check if gas concentration exceeds the safe threshold
  if (sensorValue > threshold) {
    digitalWrite(alertPin, HIGH); // Turn ON buzzer/LED
    Serial.println("--- ALERT: Gas Detected! ---");
  } else {
    digitalWrite(alertPin, LOW); // Turn OFF buzzer/LED
  }

  // 1-second delay before the next reading
  delay(1000);
}

Done compiling.

Sketch uses 2334 bytes (7%) of program storage space. Maximum is 32256 bytes.
Global variables use 236 bytes (11%) of dynamic memory, leaving 1812 bytes for local variables. Maximum is 2048 bytes.

20 Arduino Uno
32°C Mostly cloudy 11:36 AM 8/8/2026
```

